Fiscal Year:	FY 2009	**Task Last Updated:**	FY 12/03/2009
PI Name:	Ware, Jeffrey Hart Ph.D.		
Project Title:	The Role Of Free Radicals In Space Radiation-Induced Malignant Transformation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-6072	**Congressional District:**	2
Comments:	PI is deceased as of 2011.		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
Start Date:	09/01/2004	**End Date:**	08/31/2009
No. of Post Docs:	3	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes then, per JSC/A. Chu-ARC (jvp 4/2009) NOTE: Received NCE to 8/31/2009 per A. Chu/ARC (12/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA04CK46G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This research program investigated the role of oxidative stress in space radiation induced carcinogenesis in vitro and evaluated the efficacy of antioxidants as countermeasures against space radiation induced malignancy. The main hypothesis to be tested was that space radiation induces malignant transformation by increasing oxidative stress, and that prevention of oxidative stress during and after the radiation exposure can prevent down-stream events, thereby suppressing space radiation induced carcinogenesis. The radiation sources used were high energy protons and high-energy, highly charged particles known as HZE particles, which are the main components of space radiation of concern for the health of astronauts. The cell system used throughout the study was the HTori-3 human thyroid epithelial cell line that is non-malignant but can be transformed by ionizing radiation to yield cell clones capable of anchorage independent growth in soft agar and tumor formation in athymic nude mice. The study plan involved overespression of four major antioxidant enzymes, i.e., manganese superoxide dismutase (MnSOD), copper-zinc superoxide dismutase (CuZnSOD), catalase (CAT) and glutathione peroxidase (GPx), in HTori-3 cells. These cells were then exposed to proton and HZE particle radiation in the presence or absence of small molecular weight antioxidants to determine whether the over-expressed antioxidant enzymes protected the cells against space radiation induced malignant transformation and whether the small molecular weight antioxidants can substitute for antioxidant enzymes or act in synergy with the antioxidant enzymes to protect cells against malignant transformation. These studies are expected to yield valuable information to ascertain the role of oxidative stress in space radiation induced carcinogenesis and the feasibility of modulating antioxidant enzyme activity and/or diet supplementation with small molecular weight antioxidants as countermeasures against space radiation induced malignancy.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the role of oxidative stress in carcinogenesis--and methods of modulating carcinogenesis rates by modulating oxidative stress--can yield great benefits in the prevention of cancer in those exposed to elevated levels of oxidative stress. Such situations may include people exposed to ionizing radiations such as victims of radiation terrorism (i. e., a dirty bomb) and patients receiving radiation therapy, as well as people exposed chemical oxidants (e. g., smokers.)
Task Progress:	1. Cell line establishment. The first part of this grant work involved generating HTori-3 derivative cell lines. We did this using a neomycin-resistant selectable marker and obtained HTori-3 cells stably expressing the antioxidant enzymes manganese superoxide dismutase (MnSOD), copper-zinc superoxide dismutase (CuZnSOD), catalase (CAT) and glutathione peroxidase (GPx). In addition, we also established clones stably transfected with multiple antioxidant genes. One of these is HTori-3 cells transfected with CuZnSOD and CAT. The other clone is HTori-3 cells transfected with MnSOD and GPx. Overexpression of these enzymes was confirmed by Northern blot. Increased biochemical activity in transfected cells was confirmed by biochemical assays on cell extracts and by zymogram gels. In addition to these, we established two cell lines expressing the neomycin marker but not containing sequences for antioxidant genes. These were used as controls. Initial work with our control showed substantial cytological differences from the parental cell line. While this is not unexpected, we decided to establish a second clone with the empty neomycin-resistance vector. With these nine cell lines, we had established a series of cell lines with which to test our hypotheses. Because phenotypic drift of such derived cell lines is known to occur, there is a need to expand the cell lines and freeze down a large number of aliquots. 2. Cell line characterization. Transfected cell lines were characterized by measuring doubling time, plating efficiency and yield of colonies in soft agar. While plating efficiency and growth rate were somewhat variable, these differences were not deemed to be problematic in conducting the planned experiments. All transfected cell lines, including the controls, yielded fewer anchorage-independent colonies than the control cell line, whose yield was already small. 3. Radiation Response Parental and transfected HTori-3 cells were characterized by their response to gamma radiation. All cell lines produced kill curves with close fits ($R2>0.90$) to the expected exponential decay. Surviving fractions of cells exposed to 6.0 Gy. There are two different cell lines represented in this figure which are transfected only with the control vector, and these have different sensitivity to radiation at 6.0 Gy ($p<0.001$ by t-test). In addition, both of these have different sensitivities compared to the parental cell line ($p<0.001$ by t-test). Depending on which line is selected for comparison as the control, the cell lines transfected with antioxidant enzymes can be observed to be more or less sensitive to radiation. If the least sensitive control is selected (Neo#1), then no increase in survival is seen in any cell line expressing antioxidant enzymes. If the most sensitive control is selected (Neo#2), the greatest increase in surviving fraction (in HT-GPx) is increased by less than a factor of 2. Since these cell lines show increases of 5-10 fold in enzymatic activity over the control cells, we conclude that there is no biologically relevant protection from increased expression of antioxidant enzymes in this system. Some experiments were performed examining the cell killing by 1.0 GeV/n Fe. These also showed no significant differences in cell survival between parental cells and transfected cells. 4. Radiation induced transformation While antioxidants failed to protect cells from radiation induced death, the central question of this grant work was whether antioxidants could protect cells from radiation induced malignant transformation. We used yield of anchorage-independent colonies to assess induction of malignancy. To perform these assays, cultures of cells were irradiated and cultured for 1 week. The cultures were then trypsinized and the single-cell suspensions counted to measure cell concentration. Measured numbers of cells from each suspension were then plated on soft agar to measure anchorage-independent colony formation. At the same time, portions of the same cell suspensions were also plated on conventional tissue culture plastic. These were cultured in similar fashion as that used for a survival assay (10-14 days), and then stained and colonies counted to measure the proportion of viable cells in the suspension. The results of these experiments produced a similar pattern to that which we saw in the survival data; different values for the two vector controls , and different results depending on which is selected for comparison. The yield of colonies was

	then adjusted for the number of viable cells plated, which gives the clearest data on the number of cells malignantly transformed. In all cases, the yield of anchorage independent colonies from cell lines transfected with the control vector is lower than from cell lines transfected with vectors encoding antioxidant enzymes. We conclude that transfection with antioxidant enzymes confers no protection from radiation-induced malignant transformation in this system. The implications of this will be discussed at length in our publications of these data.
Bibliography Type:	Description: (Last Updated: 12/08/2009)
Abstracts for Journals and Proceedings	Ware JH, Xu Y, Zhou Z, St Clair DK, Kennedy AR. "Establishment and characterization of human thyroid cell lines overexpressing antioxidant enzymes." NASA Human Research Program Investigators' Workshop, League City, Texas, February 5, 2008. NASA Human Research Program Investigators' Workshop, League City, Texas, February 5, 2008. , Feb-2008
Abstracts for Journals and Proceedings	Ware JH, Kennedy AR. "Overexpressing antioxidant enzymes may provide no extra protection against the cell killing effects of radiation." 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, Pennsylvania, June 30, 2008. 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, Pennsylvania, June 30, 2008. , Jun-2008